Sun Devil Braking Co.

Phase 3 Report: Car Braking System

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Introduction

For phase three of this report, the main objective was to develop our design into a physical build by 3D printing, assembling, and testing the braking system to see if it was designed correctly in the previous two phases. The primary components fabricated included the rotor, caliper, brake pads, brake line, and associated pins. For a system that has many external components besides those that are in the braking system itself, it was found to be unrealistic to run an effective braking test on this system due to the system not being capable of replicating real-world braking forces or thermal conditions as covered in this report. Therefore, this report focuses on the printing and interpretation of the 3D printed braking system model, demonstrating geometric functionality and assembly feasibility, and identifying design limitations and areas for improvement, as the assembly and testing of the braking system are not possible without the use of a wheel/car, which wouldn't correspond with our 3D printed model.

Fabrication Details

In regards to the fabrication details, there was a lot to consider. As it was our first time 3D printing, multiple iterations (two reprints) had to be performed in order to ensure that the model would fit correctly without any issues. The components were made using a Flashforge Adventure 5M 3D printer using black PLA filament. The print bed temperature was set to 55°C, and the spout was set to 225°C. Default print speed settings and a 15% infill grid were used. We showcased our first prototype to the Professor, and he mentioned that the 3D printing could have been better. Thus, in the next iteration, we made sure to learn about types of supports as well as rafts in order to optimize the time as well as the material that the print would take. The rotor took 7 hours to complete, and the rest of the components took only 3 hours to complete. The parts were printed at a 53% scale due to limitations of the print bed size (220 mm x 220 mm). With the rotor being the largest component of the system (326 mm diameter), we opted to scale the physical model as such, with the added benefits of shorter print times and reduced material cost. After that, we took it to the shop in order to remove any supports that would interfere with the assembly. Due to a successful job of modifying the components to meet adjustments due to scaling and printing tolerances, the entire system could be assembled without needing additional adhesive, as components fit snugly together. All that was required was the cleanup of the supports as well as the rafts.

Assembly Procedures & Challenges

The main challenge with our assembly was the fact that, without a rotating system, a wheel, and a car to connect to, there is no practical way that the full braking system could be assembled and functional. Our braking system utilizes a “floating caliper”, and thus is separate from the rotor, meaning that we would need two additional separate structures to suspend the system while also mimicking the functionality of a car’s wheel hub/axle. Furthermore, the PLA filament used for printing is a poor substitute for the metals chosen for each component's material. Not only is PLA not suitable for the stresses the system would undergo, but it does not have a high enough frictional coefficient for the friction material of the brake pads/rotor, and additionally, it would melt from all of the immense heat generated as all of the rotational kinetic energy is converted to thermal energy. As brakes are a hydraulic system, incredibly tight tolerances are required, especially for the piston, and specially designed gaskets and seals would be needed to ensure the system can withstand the pressure cycles, while also moving the piston in and out without leaking brake fluid. The tolerances of the 3D printer we used are orders of magnitude looser than what would be required. Because of this, we focused on the modeling aspects of printing. During the assembly, we encountered size errors and excess material remaining from the printing process. Due to errors during printing, the caliper was initially printed at a size too small to fit over the brake rotor. As a result, a new caliper was printed with the accurate dimensions to allow proper assembly of the braking system. In addition to this, excess material from the supports remained on the brake rotor and the caliper after the printing process. One particular instance was the raft used on the base of the brake rotor. Due to the large size of the rotor, the raft was unnecessary and ultimately could not be easily peeled away as with smaller components. This required sanding each part to ensure that everything was properly fitted and aligned once the braking system was assembled. The support structures within the internal cooling vanes of the brake rotor posed a challenge in particular, due to the narrow passageways and the inability to fit traditional tools inside. The final assembly of the model included the brake rotor, caliper, brake pads, and piston. To assemble the model, the piston was inserted into the caliper housing. One brake pad was positioned against the piston, while the second pad was placed on the opposite side of the rotor within the caliper.

The assembled caliper was then mounted over the rotor and aligned with the caliper pins, completing the partial braking system. As previously mentioned, a true physical build of the original design would require additional components to properly seal the system, as well as grease to lubricate the caliper pins, and the components would have to be realistically machined from metals instead of 3D printed with plastics.

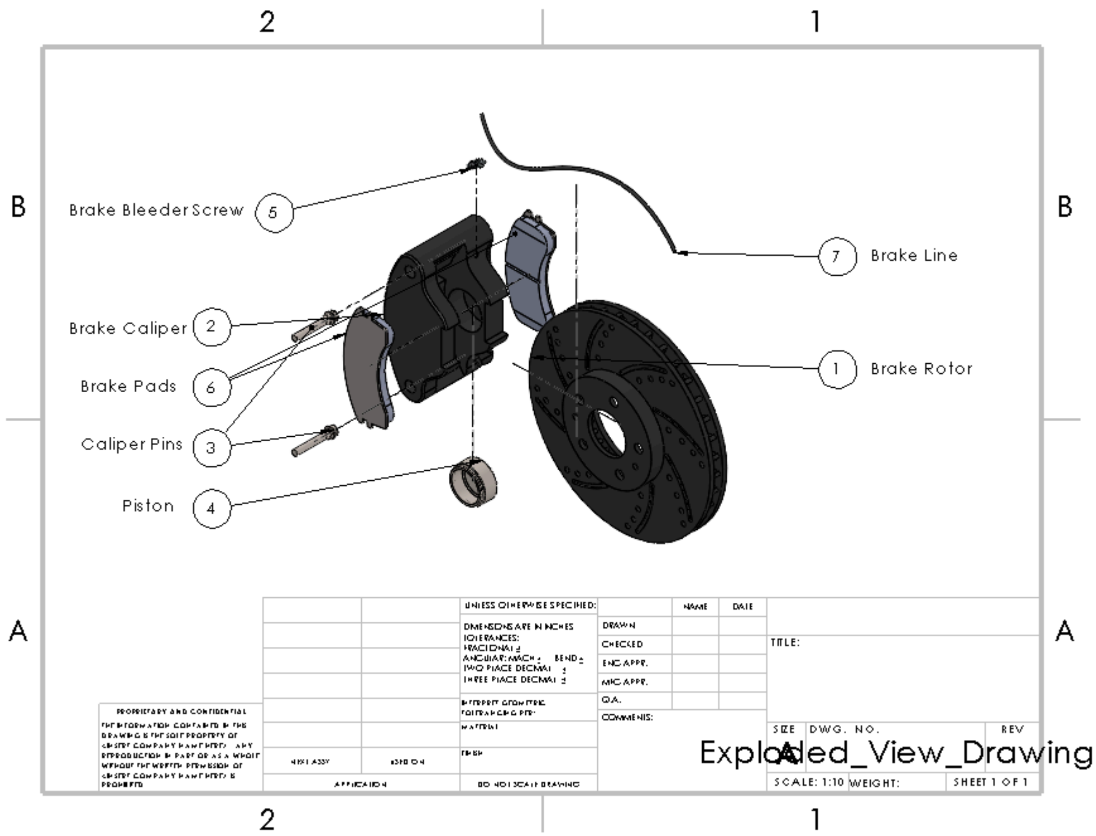

Fig. 1 - Brake System Assembly Exploded CAD Drawing

Fig. 2 - Brake System CAD Assembly

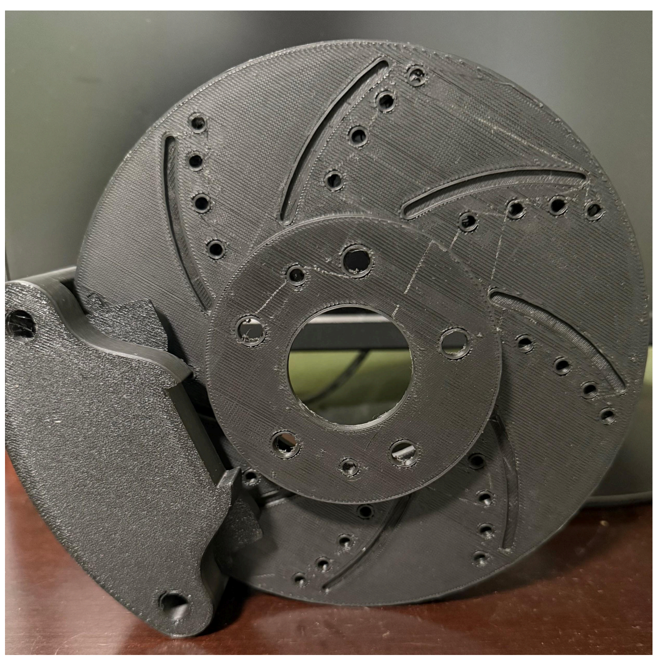

Fig. 3 - 3D Model Assembled

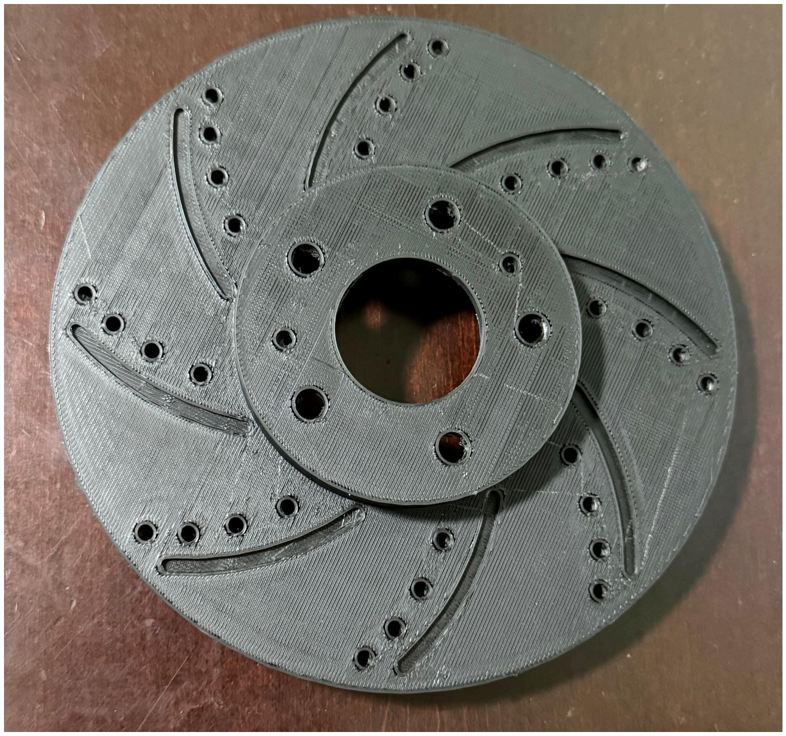

Fig. 4 - Brake Rotor

Fig. 5 - Brake Caliper

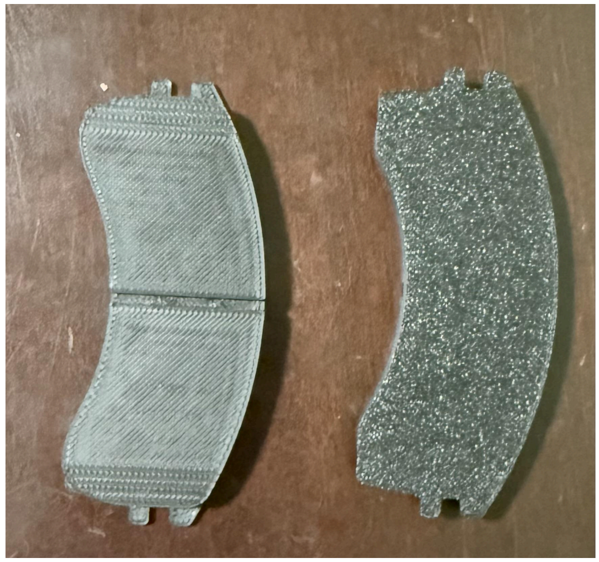

Fig. 6 - Brake pads

Fig. 7 - Piston

Test Procedures, Results, & Interpretation

A true braking test is not feasible for this 3D printed model for multiple reasons. To start, the 3D print materials are not comparable to the materials that are used in a real braking system, and the materials in a real braking system make a huge difference in pieces such as the brake rotor and brake pads that must have certain thermal and frictional properties. Therefore, a 3D printed rotor and brake pads would not provide a sufficient test to compare to a real braking system. In addition to materials, the scale is too small to generate realistic forces. Also, for this 3D printed model, there is no rotating system/vehicle available for the braking system to be connected to. Without this, there is no good way to test the braking system. Instead, we did some functional testing, such as making sure the rotor rotation manually worked and the clearance between the rotor and pads fit correctly. We also tested that the caliper could be manually compressed to simulate the braking action and see how the friction attempted to slow the motion of the rotor. Of course, as stated, the low-friction materials and lack of real clamping made the demonstration and results unrealistic and incomparable to reality. Overall, the system successfully demonstrated mechanical concepts and component interaction but could not replicate real braking physics due to material and parts limitations. If we were to repeat testing using a physical build of our braking system that resembles the design, we would be most interested in the thermal stresses produced within the brake rotor, as well as wear testing. As the brake rotor undergoes many rotations and braking cycles, failure is primarily due to deformation caused by wear or warping due to pressure while at temperatures nearing the material's melting point.

Phase 2 Comparison

Comparing this section of 3D printing and testing with the last phase, which was primarily running analysis, is slightly tricky, as the team wasn't able to accurately test the braking system due to many limitations. From the last phase, the overall geometry and component layout were successfully replicated into a physical model. Also, functional relationships such as the rotor-pad-caliper interaction were preserved, but could not be realistically tested with the physical model. Some of the key differences were the real-world assumptions we made in phase two, such as the material strength, friction coefficients, calculated pressures, rotor rotational velocity, and other factors that were considered in the analysis portion, did not translate into the model, because the model is all 3D printed with much different materials and numerical values. This affected the expected performance metrics that could be validated with the physical model, such as the force, pressure, and stopping ability. We could not validate the physical model with the analysis run in phase two. Lastly, manufacturing tolerances are different than the CAD design tolerances, which also affected the ability to validate the physical model. Overall, from phase two, the main comparable aspects are the overall geometry and layout of the braking system and the functional relationship between components.

Failures, Mistakes, Surprises

Failures:

- Unable to demonstrate the full braking system as an integrated assembly
- This was caused by the rotor and caliper being floating components with no attachment to a hub, making it impossible to mount the system in a functional context
- No hydraulic system was integrated, so the piston force required to compress the brake pads cannot be demonstrated or tested

Mistakes:

- A printing error occurred due to an unnecessary raft on the bottom, affecting surface finish when not needed

- The initial calliper was printed using a 35% model of the original, when the intended sizing was 53%

Surprises:

- Printing components in the correct orientation prevented hole misalignment and improved dimensional accuracy
- Despite the lack of hub mounting and hydraulics, the overall geometric fit and mechanical interaction turned out better than expected
- No extra adhesive or pins were needed to connect the brake pads to the calliper, as the tolerances in the CAD model allowed proper fitting even with printer inaccuracies

<u>Design Change Considerations for a Second Iteration</u>

1. **Material Improvements:** Ideally, a second iteration would require realistic materials for the brake pads and the rotor so that all of the analyses that were run on ANSYS could be used as a point of reference and comparison. This would make the friction interaction between the pads and rotor more realistic and comparable with a real braking system.
2. **Functional Testing Improvements:** Improvements would need to be made so that the braking system could actually be somewhat tested. Adding a hand crank, motor, or some type of mechanism to apply a controlled clamping force would allow braking to be simulated.
3. **Improve Alignment & Tolerances:** Depending on the printer for a second iteration, adding intentional clearances and standardized hole size for the pins would allow for

easier assembly. Guides and slots could also be added to ensure proper caliper positioning or the centering of the rotor.

4. **Surface Finish Enhancements:** The rotor and brake pads could be sanded or polished in order to achieve more realistic material friction properties
5. **Reinforce Critical Components:** To ensure the strength of the system, the thickness of pins and critical components could be increased, or the print orientation could be changed to improve the strength of the system.